

Wave 5.2R Offline Leader Cross-Surface Promotion Campaign Results

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Executive Summary

The approved campaign completed 27 / 27 runs with zero failed queue entries: 18 K01/H08 promotion runs and 9 matched H04 analytical-anchor runs. Every declared checkpoint is present.

K01 is the provisional scalar validation winner on Fw with seed 271828 and test MAE 0.001355 deg. Across the three seeds, K01 also has the lowest mean test MAE on Fw, Bw, and direction-aware global. This is campaign evidence only: no global promotion and no incumbent replacement is authorized.

Campaign Contract

- Dataset: polished_dataset.
- Input mode: setpoints.
- Surfaces: Fw, Bw, and direction-aware global.
- Seeds: 314159, 271828, and 161803.
- Promotion candidates: K01 and H08.
- Internal analytical anchor: H04.
- Runtime target-derived inputs: zero.
- Completed runs: 27.
- Failed runs: 0.

Aggregate Raw-Error And Repeatability Results

Surface	Candidate	Mean MAE [deg]	MAE SD [deg]	Best MAE [deg]
Fw	K01	0.001442	0.000088	0.001355
Fw	H08	0.001690	0.000003	0.001686
Fw	H04	0.001749	0.000040	0.001716
Bw	K01	0.001636	0.000050	0.001580
Bw	H08	0.001973	0.000028	0.001934
Bw	H04	0.001995	0.000012	0.001978
global	K01	0.001553	0.000067	0.001460
global	H08	0.001869	0.000001	0.001868
global	H04	0.001892	0.000011	0.001883

Aggregate Shape And Offset Results

Surface	Candidate	Centered MAE [deg]	Mean offset [deg]
Fw	K01	0.001243	0.000558
Fw	H08	0.001349	0.000841
Fw	H04	0.001365	0.000909
Bw	K01	0.001407	0.000548
Bw	H08	0.001683	0.000738
Bw	H04	0.001716	0.000721
global	K01	0.001317	0.000583
global	H08	0.001521	0.000835
global	H04	0.001544	0.000839

K01 leads mean raw MAE and mean centered MAE on every trained surface. H08 is highly repeatable across seeds and improves the matched H04 raw MAE on every surface, but its scalar offset advantage is surface-dependent. These aggregate training/test metrics do not replace the required per-curve robustness, harmonic, phase, continuity, and visual evidence.

Provisional Winner

- Candidate: K01 .
- Surface: Fw .
- Seed: 271828 .
- Run: 2026-07-31-10-45-41__stage9_k01__seed_271828 .
- Test MAE: 0.001354961 deg .
- Centered MAE: 0.001205198 deg .
- Offset absolute error: 0.000490818 deg .
- Per-curve MAE P95: 0.003932765 deg .
- Chunk-equivalence maximum difference: 3.948808e-07 deg .
- Reset reproducibility maximum difference: 0 deg .

The provisional winner is selected by the campaign's validation-only scalar ordering. It cannot authorize promotion under the repository curve-first policy.

Integrity And Bookkeeping

All nine queue-state files report completed , all 27 result rows have matching checkpoints, and the campaign state reports 27 completed and 0 failed. The closeout repaired a bookkeeping omission where the nine successful K01 rows lacked the CSV/YAML status field even though their queue states, metrics, predictions, and checkpoints were complete. The campaign runner now writes that field for future executions.

Incumbent Preservation

Periodic GRU remains the accepted temporal non-PINN reference. Periodic harmonic MLP remains the accepted non-temporal non-PINN reference. Neither is deleted, overwritten, or demoted by this scalar campaign closeout.

Promotion Decision

The campaign qualifies K01 and H08 for the separate official TE Curve Verification Pipeline refresh. It does not yet establish that either model is a global leader. The official review must keep `Fw` , `Bw` , and `global` separate and compare raw error, centered shape, offset and continuity, harmonic and phase fidelity, robustness, visual evidence, and deployment readiness.

Registry Decision

No accepted family or program leader registry changes in this normal closeout. K01 remains the temporal offline leader and H08 remains the balanced non-temporal offline leader pending official curve-first verification and target-runtime acceptance.

Future Integrated Specialist TODO

The roadmap retains the separate design study combining the complementary strengths of K01, H08, F01, S01, H04, Stage 10 R00, and Stage 10 S01. This closeout does not authorize that model or reopen physics-integrated Wave 6.

Reproducibility Evidence

- Campaign directory: `output/training_campaigns/2026-07-31-10-39-08_wave52r_offline_leader_cross_surface_promotion_2026_07_30` .
- Leaderboard: `campaign_leaderboard.yaml` .
- Provisional winner: `campaign_best_run.yaml` and `campaign_best_run.md` .
- Result table: `campaign_results.csv` .
- Immutable run list: `campaign_artifact_path_list.txt` .

Next Step

Prepare the operator-facing `Fw` / `Bw` / `global` TE Curve Verification Pipeline launcher, run it separately, and accept or reject each candidate surface using the official multi-index curve-first policy.